Project Analysis & Evaluation

Risk, Break-Even, and Capital Rationing

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Learning Objectives

1. Scenario Analysis
2. Sensitivity Analysis
3. Break-Even Analysis
4. Degree of Operating Leverage (DOL)
5. Capital Rationing
6. Illustrative Example: Project Overview
7. Illustrative Example: Base Case Analysis
8. Illustrative Example: Scenario Analysis
9. Illustrative Example: Break-Even
10. Illustrative Example: Operating Leverage
11. Illustrative Example: Capital Rationing

1. Scenario Analysis

- **Definition:** Analyzing the impact on NPV when multiple variables (Sales, Price, Costs) are changed at the same time.
- **Purpose:** To determine the range of possible outcomes (Worst to Best).

Illustrative Case (Base Case)

Initial Investment: $200,000 — Project Life: 5 Years — Discount Rate: 10%.

- Unit Sales: 6,000 — Price: $80 — Variable Cost: $60
- Fixed Costs: $50,000

Example: Scenario Results

Variable	**Worst Case**	**Base Case**	**Best Case**
Unit Sales	4,500	6,000	7,500
Price per Unit	$75	$80	$85
Variable Cost	$62	$60	$58
NPV	**-$35,200**	**$79,500**	**$215,000**

Interpretation:

- The project is risky because the **Worst Case** results in a negative NPV.
- Management must evaluate the probability of the Worst Case scenario.

2. Sensitivity Analysis

- **Definition:** Investigating the effect on NPV when **only one** variable is changed while others stay constant (*ceteris paribus*).
- **Goal:** Identify the "Critical Variable" (the biggest risk driver).

Illustration: Price Sensitivity

What if only the Price drops by 10% (from \$80 to \$72)?

- New OCF drops significantly → NPV falls to \$25,000.
- If Unit Sales dropped 10% instead, NPV would only fall to \$65,000.
- **Conclusion:** The project is more sensitive to **Price** than to Volume.

Comparison: Scenario vs. Sensitivity Analysis

Feature	Scenario Analysis	Sensitivity Analysis
Variables Changed	Multiple variables simultaneously.	One variable at a time.
Primary Goal	Estimate range of outcomes (Best/Worst).	Identify the critical risk factor.
Correlation	Considers interactions between factors.	Ignores correlations.

3. Break-Even Analysis

How low can sales fall before we stop making money?

1. **Accounting Break-Even**: Net Income = 0.

$$Q = \frac{FC + D}{P - v}$$

2. **Cash Break-Even**: Operating Cash Flow (OCF) = 0.

$$Q = \frac{FC}{P - v}$$

3. **Financial Break-Even**: NPV = 0.

$$Q = \frac{FC + OCF^*}{P - v}$$

**OCF* *is the cash flow required to recover the initial investment at the required rate of return.*

Example: Break-Even Calculation

Data: Invest: \$200k, $D = \$40k$, $FC = \$50k$, Contribution $(P - v) = \$20$.

- **Accounting BEP:**
  $Q = (50,000 + 40,000)/20 = \mathbf{4,500}$ units
- **Cash BEP:**
  $Q = 50,000/20 = \mathbf{2,500}$ units
- **Financial BEP (at 10% return over 5 years):**
  Required $OCF^* = 200,000/3.7908 = \$52,760$
  $Q = (50,000 + 52,760)/20 = \mathbf{5,138}$ units

4. Degree of Operating Leverage (DOL)

- **Definition:** Measures the relationship between percentage change in OCF and percentage change in sales.
- **Formula:**

$$DOL = 1 + \frac{FC}{OCF}$$

Illustration of DOL

If $FC = \$50,000$ and $OCF = \$70,000$:

$$DOL = 1 + \frac{50,000}{70,000} = \mathbf{1.71}$$

Effect: A **1% increase** in Sales leads to a **1.71% increase** in OCF. Higher fixed costs result in higher DOL and higher business risk.

5. Capital Rationing

- **Definition:** Limited capital prevents a firm from taking all positive NPV projects.
- **Tool: Profitability Index (PI)**.

$$PI = \frac{PV \text{ of Future Cash Flows}}{\text{Initial Investment}}$$

Decision Rule

Rank projects by PI from highest to lowest. Select projects until the budget is exhausted to maximize **Total NPV**.

Example: Project Selection (Budget $100,000)

Project	Investment	NPV	PI	Rank
A	$50,000	$20,000	1.40	2
B	$40,000	$25,000	1.62	1
C	$60,000	$18,000	1.30	3

Decision:

1. Accept **Project B** (Invest $40k, $60k remaining).
2. Accept **Project A** (Invest $50k, $10k remaining).
3. **Reject Project C** (Not enough budget, despite positive NPV).
4. **Total NPV Realized: $45,000**.

ILLUSTRATIVE EXAMPLE

Case Study: The Sugarcane Plant

Evergreen Bio-Industries is evaluating a new processing facility.

- **Initial Investment:** \$10,000,000 (10-year Straight-Line Depreciation).
- **Sales Volume:** 200,000 tons/year.
- **Price per Ton:** \$500 — **Variable Cost:** \$350.
- **Fixed Operating Costs:** \$1,500,000/year.
- **Financials:** 12% Required Return (r), 21% Tax Rate (t).

Objective

Determine project viability, risk sensitivity, and survival thresholds.

Step 1: Base Case OCF and NPV

1. Annual Depreciation (D):
$D = \$10,000,000/10 = \$1,000,000$

2. Operating Cash Flow (OCF):
$EBIT = (Sales - VC - FC - D)$
$EBIT = (100M - 70M - 1.5M - 1M) = \$27,500,000$
$OCF = EBIT(1 - t) + D$
$OCF = 27.5M(0.79) + 1M = \$\mathbf{22,725,000}$

3. Base Case NPV:
$NPV = -Initial + (OCF \times PVIFA_{12\%,10y})$
$NPV = -10M + (22,725,000 \times 5.6502) = \$\mathbf{118,400,795}$

Step 2: Scenario Analysis (The "Sugar Crash")

What happens if the market faces a surplus and supply chain issues?

Variable	**Base Case**	**Worst Case**
Price per Ton	\$500	\$400
Variable Cost	\$350	\$380
EBIT	\$27,500,000	\$1,500,000
OCF	\$22,725,000	\$2,185,000
NPV	**\$118,400,795**	**\$2,345,687**

Interpretation: Even in a high-stress scenario, the project maintains a positive NPV, indicating **robust profitability**.

Step 3: Financial Break-Even Analysis

To find the quantity (Q) where $NPV = 0$, we first find the required OCF^*:

$$OCF^* = \frac{Investment}{PVIFA} = \frac{10,000,000}{5.6502} = \$1,769,849$$

Solving for Q:

$$Q = \frac{FC(1-t) + OCF^* - D(t)}{(P-v)(1-t)}$$

$$Q = \frac{1,500,000(0.79) + 1,769,849 - (1,000,000 \times 0.21)}{(500-350)(0.79)}$$

$$Q = \frac{2,744,849}{118.5} = \mathbf{23,163} \text{ tons}$$

The plant must operate at ˜11.6% capacity to break even financially.

Step 4: Degree of Operating Leverage (DOL)

DOL measures the "multiplier effect" of sales on cash flow.

$$DOL = 1 + \frac{FC}{OCF_{pre-tax}}$$

$$DOL = 1 + \frac{1,500,000}{28,500,000} = \mathbf{1.05}$$

Analysis

A DOL of 1.05 is **very low**. This suggests the project has a low proportion of fixed costs relative to its massive revenue.

- **Risk:** A 10% drop in sales volume only leads to a 10.5% drop in OCF.
- **Safety:** High margin of safety.

Step 5: Capital Rationing (Budget: $15M)

The firm must choose between the **Sugarcane Plant** and an **Ethanol Distillery**. They are mutually exclusive due to a $15M budget limit.

Project	Investment	NPV	PI
Sugarcane Plant	$10,000,000	$118,400,000	12.84
Ethanol Distillery	$7,000,000	$90,000,000	**13.85**

The Conflict:

- **Profitability Index (PI):** Ethanol is more "efficient" (higher PI).
- **NPV Maximization:** Sugarcane adds more total value to the firm.

Conclusion

Under **Hard Rationing**, choose the **Sugarcane Plant** to maximize the total wealth created ($118.4M vs $90M).

Illustrative Example: Summary

- **Viability:** Extremely strong Base Case NPV of $118.4M.
- **Sensitivity:** Critical factor is the price of sugar; however, the project survives even with a 20% price drop.
- **Operating Risk:** Low (DOL 1.05).
- **Strategic Choice:** Superior to the Ethanol Distillery in total value creation despite a slightly lower PI.

Recommendation: ACCEPT PROJECT

Thank You

Any Questions?